Keep in Mind

This is a practice test. It mirrors the real Unit 4 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Rational function ☐
2. Domain restriction ☐
3. Vertical asymptote ☐
4. Hole ☐
5. Horizontal asymptote ☐
6. Least common denominator ☐
7. Extraneous solution ☐
8. Constant of variation ☐

(A) A value that makes the *original* denominator zero, so it can never be an input.

(B) A horizontal line the graph approaches at its far left and far right; found by comparing the degrees of the numerator and denominator.

(C) The number k in $y = kx$ or $y = \frac{k}{x}$; you find it from one known pair of values.

(D) A quotient of two polynomials whose denominator is not zero.

(E) A single missing point on an otherwise unbroken curve, left behind by a factor that cancels.

(F) A number that turns up when you solve but must be thrown out, because it is an excluded value of the original equation.

(G) The smallest expression that every denominator divides evenly, built from the factored denominators.

(H) A vertical line the graph runs alongside but never touches, left by a factor that does *not* cancel.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. The domain of $f(x) = \dfrac{x+5}{x^2-9}$ is all real numbers except (A) $x = -5$ (B) $x = 3$ (C) $x = \pm 3$ (D) none — all reals **Letter:** ☐

2. Simplified completely, $\dfrac{x^2-25}{x^2+3x-10}$ is (A) $\dfrac{x-5}{x-2}$, $x \neq 2$ (B) $\dfrac{x-5}{x-2}$, $x \neq -5, 2$ (C) $\dfrac{x+5}{x-2}$, $x \neq 2$ (D) $\dfrac{-5}{3x-10}$ **Letter:** ☐

3. The horizontal asymptote of $f(x) = \dfrac{3x^2-1}{x^2+4x}$ is ☐ (A) $y = 0$ (B) $y = \frac{1}{3}$ (C) $y = 3$ (D) there is none

4. Which statement about $h(x) = \dfrac{x^2-x-6}{x^2-9}$ is true? **Letter:** ☐
   (A) There are vertical asymptotes at $x = 3$ and at $x = -3$.
   (B) There is a hole at $x = 3$ and a vertical asymptote at $x = -3$.
   (C) There is a hole at $x = -3$ and a vertical asymptote at $x = 3$.
   (D) There is a hole at $x = -2$ and a vertical asymptote at $x = 3$.

5. The graph of $g(x) = \dfrac{1}{x-4} + 2$ has asymptotes (A) $x = 4$, $y = 2$ (B) $x = -4$, $y = 2$ (C) $x = 4$, $y = 0$ (D)

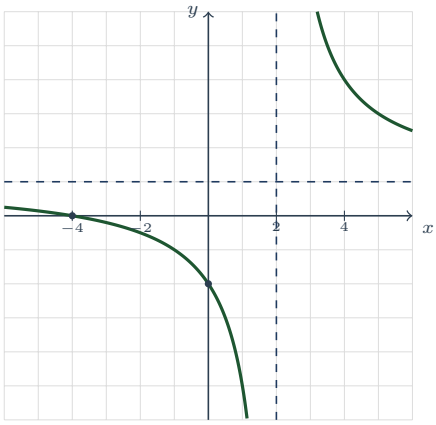
$x = 2,\ y = 4$ **Letter:** ☐

6. y varies *inversely* as x, and $y = 8$ when $x = 3$. When $x = 12$, $y =$ ☐ (A) 2 (B) 6 (C) 24 (D) 32

Part C — Short Answer & Computation (40 pts)

Show your work. State every restriction on the variable.

1. **Read the graph.** The rational function f is graphed below; the dashed lines are its asymptotes and each gridline is one unit.



(a) Equation of the vertical asymptote: ☐ Equation of the horizontal asymptote: ☐

(b) x-intercept: ☐; y-intercept: ☐

(c) Domain (interval notation): ☐

(d) Range (interval notation): ☐

(e) End behavior: as $x \to \pm\infty$, $f(x) \to$ ☐

(f) Interval(s) where f is decreasing: ☐

(g) Interval(s) where $f(x) > 0$: ☐

2. **Simplify and state the restrictions.** Read the restrictions from the *original* denominator.

(a) $\dfrac{3x^2 - 27}{x^2 + x - 12}$ (b) $\dfrac{x^2 - 16}{4 - x}$

(a) ☐ $x \neq$ ☐ **(b)** ☐ $x \neq$ ☐

3. **Multiply or divide.** Factor first; give the simplest form with its restrictions.

(a) $\dfrac{x^2 - 4}{x^2 + 6x + 9} \cdot \dfrac{x + 3}{x - 2}$

Answer: ☐ $x \neq$ ☐

(b) $\dfrac{x^2 - x - 6}{x^2 - 1} \div \dfrac{x - 3}{x + 1}$

Answer: $\square$ $x \neq$ $\square$

4. **Add, subtract, and simplify a complex fraction.**

   (a) $\dfrac{5}{x+2} - \dfrac{3}{x-1}$

   Answer: $\square$ $x \neq$ $\square$

   (b) $\dfrac{x}{x^2-9} + \dfrac{2}{x+3}$

   Answer: $\square$ $x \neq$ $\square$

   (c) $\dfrac{\frac{1}{x}+\frac{1}{2}}{\frac{1}{x}-\frac{1}{2}}$

   Answer: $\square$ $x \neq$ $\square$

5. **The parent function and its transformations.** Let $g(x) = \dfrac{1}{x+3} - 2$.

   (a) Describe the transformation(s) that take $y = \dfrac{1}{x}$ to g. $\square$

   (b) Vertical asymptote: $\square$; horizontal asymptote: $\square$

   (c) Domain: $\square$; range: $\square$

   (d) A different hyperbola of the form $y = \dfrac{1}{x-h} + k$ has vertical asymptote $x = 1$ and horizontal asymptote $y = 4$. Write its equation. $\square$

6. **Analyze from the equation.** For $f(x) = \dfrac{2x^2-8}{x^2-x-6}$:

   (a) Factor the numerator and denominator, and list every excluded value.

   Excluded: $\square$ **(b) Hole:** $\square$

   (b) Which excluded value gives a *hole* and which gives a *vertical asymptote*? Give the coordinates of the hole.

   (c) Horizontal asymptote: $\square$ (Justify with a degree comparison.)

   (d) x-intercept: $\square$; y-intercept: $\square$

7. **Solve. Check every answer against the excluded values.**

   (a) $\dfrac{x}{x+2} + \dfrac{1}{x-1} = 1$

Solution:

(b) $\dfrac{x}{x-4} = \dfrac{4}{x-4} + 3$

Solution:

(c) $\dfrac{x}{x-2} + \dfrac{2}{x} = \dfrac{4}{x^2-2x}$

Solution:

8. **Variation.** The table below records a pair of related quantities.

x	2	4	5	10
y	30	15	12	6

(a) Is y directly or inversely proportional to x? Give the computation from the table that proves it.

(b) Find the constant of variation k and write the equation.

(c) Use your equation to find y when $x = 8$.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \dfrac{x^2 - 5x + 4}{x^2 - 16}$. **(a)** Factor both parts and list every excluded value of the *original* function. **(b)** Run the cancel test: which excluded value is a hole and which is a vertical asymptote? Give the hole's coordinates and explain how you found its height. **(c)** State the horizontal asymptote, justify it with a degree comparison, and write the end behavior it describes. **(d)** Give the x- and y-intercepts. **(e)** Write the domain in interval notation. **(f)** A classmate simplifies f to $\dfrac{x-1}{x+4}$ and says the two functions are the same. Say exactly where they agree and where they do not.

2. **Modeling.** A club pays a one-time setup fee of \$180 to a screen printer, plus \$7 for each shirt printed. The *average cost per shirt* for an order of n shirts is

$$A(n) = \frac{7n + 180}{n}.$$

(a) Explain why A is a rational function, and state the domain that makes sense for this situation. **(b)** Find $A(20)$ and $A(60)$, with units. What is happening to the average cost? **(c)** Rewrite $A(n)$ in the form $7 + \dfrac{180}{n}$ and use it to state the horizontal asymptote. What does that number mean about this order? **(d)** Find the order size for which the average cost is exactly \$9 per shirt. **(e)** Can the average cost ever be exactly \$7? Answer using the horizontal asymptote.

Part D, item 1 (The full analysis)